IILM UNIVERSITY

School of Computer Science & Engineering | Greater Noida, U.P.

AICTE & ICAC APPROVED INTERNSHIP

Image Recognition System Using Machine Learning Classification Techniques

An internship presentation report on image classification using Naive Bayes and Support Vector Machine (SVM) algorithms

YASH DIXIT

Roll No. 25SCS1003004241 | B.Tech CSE | Batch 2026–27

Internship Duration: 21 June 2026 – 23 July 2026 (1 Month)

Organization: Codec Technologies Pvt. Ltd.

Candidate & Program Overview

Candidate Details

STUDENT NAME

YASH DIXIT

ROLL NUMBER

25SCS1003004241

DEGREE

B.Tech (Computer Science & Engineering)

ACADEMIC BATCH

2026 – 2027

UNIVERSITY

IILM University, Greater Noida, U.P.

Internship Profile

ORGANIZATION

Codec Technologies Pvt. Ltd.

PROGRAM

AICTE & ICAC Approved Internship Program

SPECIALIZATION DOMAIN

Artificial Intelligence — Image Recognition

DURATION

1 Month (21 Jun 2026 – 23 Jul 2026)

PROGRAM PARTNER

Google for Education / National Internship Portal

Organization Profile: Codec Technologies Pvt. Ltd.

Codec Technologies Pvt. Ltd. is a technology-focused organization that runs industry-oriented internship programs in collaboration with the National Internship Portal, under AICTE and ICAC approval, and in partnership with Google for Education — giving students practical, project-based learning experience in AI, ML, Python development and Data Science.

AICTE & ICAC Collaboration

Operates under the patronage of the AICTE National Internship Portal (ICAC approved) to empower engineering talent across India.

Industry Alignment

Partners with Google for Education to bridge the gap between academic learning and real-world AI/ML industry practice.

Core Focus Domains

Artificial Intelligence, Machine Learning, Python Development, Computer Vision and Data Science.

Problem Statement & Project Scope

The Challenge

Manual identification and categorization of images is time-consuming, inconsistent, and does not scale as image data grows. Applications such as quality inspection, document sorting, medical-imaging support and object cataloguing need automated classification based on visual features — colour, shape, texture and edges.

The core problem: how can a system automatically extract meaningful visual features and classify images accurately and consistently, without manual intervention?

The Approach & Scope

The project builds a classical machine-learning image classification pipeline — image acquisition, preprocessing, feature extraction, model training and evaluation — using Naive Bayes and Support Vector Machine (SVM) classifiers.

It is a learning-oriented prototype demonstrating the full image-recognition workflow, and is scoped to classical ML rather than deep learning, with room to extend using CNNs for higher accuracy on larger datasets.

Core Project Objectives

Build the System

Develop an Image Recognition System in Python to classify images by their visual features.

Preprocess Images

Resize, normalize and denoise raw image data to prepare it for feature extraction.

Extract Features

Extract relevant visual features from images using computer vision techniques.

Train Classifiers

Train Naive Bayes and SVM models to classify images into correct categories.

Evaluate Performance

Evaluate and compare classifier performance using standard accuracy metrics.

Gain Experience

Gain practical experience applying AI/ML to a real, end-to-end classification pipeline.

Technologies & Tools Used

PROGRAMMING LANGUAGE

Python

CORE LIBRARIES

NumPy, Pandas, Matplotlib, Scikit-learn

IMAGE PROCESSING

OpenCV, Pillow (PIL)

ML ALGORITHMS

Naive Bayes, Support Vector Machine (SVM)

DEVELOPMENT ENVIRONMENT

Jupyter Notebook / Google Colab

VERSION CONTROL

Git and GitHub

System Architecture

A linear five-stage pipeline takes a raw image through to a predicted class label:



Each stage was developed, tested and debugged independently before integration into the final system.

Methodology

1

Data Collection

Gathered a labelled image dataset covering the target classification classes.

2

Image Preprocessing

Resized images to a fixed dimension, unified colour format, normalized pixel values.

3

Feature Extraction

Extracted discriminative colour, texture and shape-based descriptors.

4

Train-Test Split

Divided the dataset into training and testing subsets for unbiased evaluation.

5

Model Training

Trained Naive Bayes and SVM classifiers on the extracted feature vectors using Scikit-learn.

6

Model Evaluation

Evaluated both classifiers using accuracy, precision, recall and confusion matrices.

7

Result Analysis

Analysed misclassifications and documented possible improvements.

Expected Outcomes

- A working Python-based Image Recognition System capable of classifying images into correct categories.
- Comparative performance results between the Naive Bayes and SVM classifiers on the same dataset.
- A clear, documented understanding of the complete image classification pipeline — from raw image to predicted label.
- Practical, hands-on experience with image preprocessing, feature extraction and classical ML techniques in Python.
- A reusable codebase that can later be extended with deep learning approaches such as CNNs for improved accuracy.

Certification & Program Verification

A copy of the Internship Completion Certificate issued by Codec Technologies Pvt. Ltd. has been attached in the full report.

Verification Metric	Details
Internship Title	Image Recognition System — Artificial Intelligence Internship
Issuing Organization	Codec Technologies Pvt. Ltd.
Academic / Program Patronage	AICTE & ICAC (National Internship Portal)
Role	Python Developer Intern — Artificial Intelligence Track
Duration	21 June 2026 – 23 July 2026 (1 Month)
University	IILM University, Greater Noida, U.P.